

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes the following points: • (organ is a) group of tissues {working together / perform similar functions} (1) • which (forms a discrete unit that) can {detect / sense} a {stimulus / change in the environment} (1)	ignore cells accept respond to a {stimulus / change in environment} accept contains nerve endings accept detect sensations ignore meanings of 'sense' organ ignore contains receptors	(2)

Question number	Answer	Additional guidance	Mark												
8(b)	Choose an item. <table border="1"> <thead> <tr> <th>Connective tissue</th><th>Contains collagen</th><th>Rich in nerves</th><th>Envelops muscle</th></tr> </thead> <tbody> <tr> <td>fascia</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>ligaments</td><td>✓</td><td>x</td><td>x</td></tr> </tbody> </table>	Connective tissue	Contains collagen	Rich in nerves	Envelops muscle	fascia	✓	✓	✓	ligaments	✓	x	x	All correct 2 marks 3 to 5 correct = 1 mark 0 to 2 correct = 0 marks accept yes for ✓ accept no for x ignore blank boxes	(2)
Connective tissue	Contains collagen	Rich in nerves	Envelops muscle												
fascia	✓	✓	✓												
ligaments	✓	x	x												

Question number	Answer	Additional guidance	Mark
8(c)	An answer that includes three of the following points: - stimulus detected by {specialised pain receptors / nociceptors} in fascia (1) - causes release of neurotransmitters / chemical response (1) - which causes {nerve impulse / action potential / depolarisation} (in sensory neurones) (1) (impulse) transmitted to pain centre (in brain / CNS) (1)	accept fascia rich in {nerves / nerve endings} accept fascia is a sensory organ accept increase in percentage of {nociceptive fibres / pain receptors (that respond to harmful stimuli)} accept causes inflammation accept opening of sodium ion channels / sodium ions enter (sensory) neurone accept {brain / CNS} interprets the impulse as pain	(3)

Question number	Answer	Additional guidance	Mark
8(d)	A description that includes three of the following points: • {tissue injury / cell damage / bacterial or viral infection / foreign antigen / exercise} (1) • release of {histamine / cytokines / hormone / chemical} (1) • (therefore) vasodilation (of fascia blood vessels) occurs (1) • {swelling / oedema} occurs due to {increase / entry} of {macrophages / white blood cells / fluid} (in the fascia) (1)	accept injury / harm / damage (to fascia) unqualified ignore fibromyalgia ignore adrenaline accept increased permeability of capillaries accept description of vasodilation / increased blood flow accept entry of {white blood cells / macrophages} (from blood)	(3)

Question number	Answer	Additional guidance	Mark
8(e)	A description that includes four of the following points: - fibroblasts respond to {mechanical / chemical stimuli / TGF-beta / transcription factor} (1) - causing {a change in gene expression / switching genes on / differential gene expression} (1) - due to {epigenetic modification / histone modification / DNA methylation} (1) {active / switched on} genes are transcribed (into mRNA) (1) - translation (of mRNA) occurs (at the ribosome) (1) (resulting in) formation of proteins needed for {differentiation / specialisation} {of fibroblasts / to myofibroblasts} (1)	accept fibroblasts 'drink' TGF-beta accept TGF-beta enter cell accept genes are activated accept formation of proteins needed for myofibroblast formation e.g collagen accumulation	(4)

Question number	Answer	Additional guidance	Mark
8(f)	An answer that includes the following points: • lymph flows due to {external mechanical compressions / contraction of (skeletal) muscles / peristalsis / contraction} of (lymph) vessel (1) • (one direction due to) valves in the lymphatic system (1) • {stimulation / coordination} by {parasympathetic / sympathetic} nervous system (1)	Accept muscle contractions unqualified accept cytoplasmic extent of endothelial cells	(3)

Question number	Answer	Additional guidance	Mark
8(g)	An answer that includes three of the following points: • lymphatic vessel walls become {thinner / weaker} (1) • due to {less / thinner / weaker} muscle (in lymphatic vessels) (1) • {less / thinner / weaker} elastic tissue (in lymphatic vessels) (1) • deterioration of {nerve network / nerves} (in endothelial layer) (1) • decrease in number of {(lymph or blood) vessels / lymphangions} (1)	ignore blood accept lymph for lymphatic vessel accept aneurysms develop (in the lymph) ignore reduced {contraction of muscle / peristalsis} accept less elasticity	(3)

